Lab Documentation: Programs 29 & 30

Program 29: Bar Chart with Unique Colors

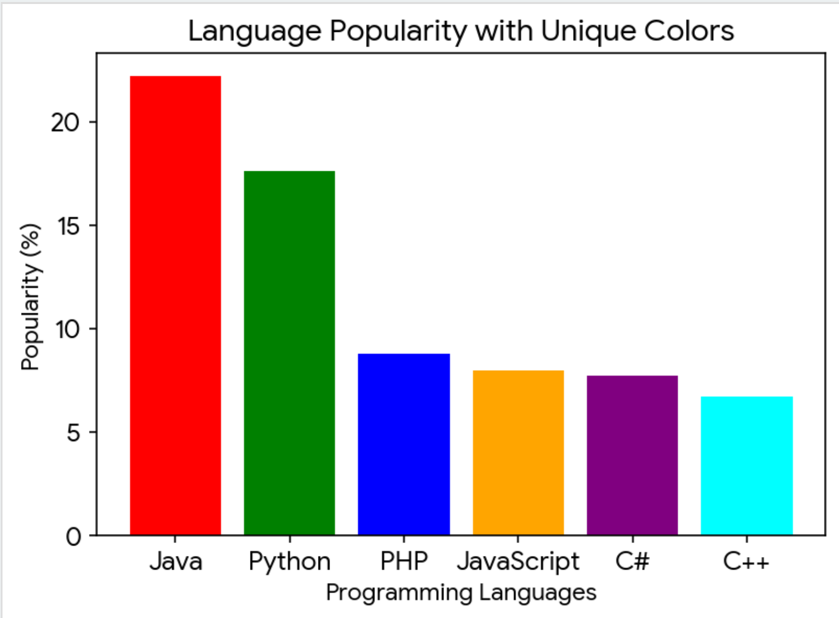
AIM

To write a Python program to display a bar chart of popularity of programming languages, using a different color for each bar.

CODE

```
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
colors = ['red', 'green', 'blue', 'orange', 'purple', 'cyan']

plt.bar(languages, popularity, color=colors)
plt.xlabel('Programming Languages')
plt.ylabel('Popularity (%)')
plt.title('Language Popularity with Unique Colors')
plt.show()
```

OUTPUT



Result: The multi-colored bar chart was generated successfully to visually distinguish each category.

Program 30: Grouped Bar Plot

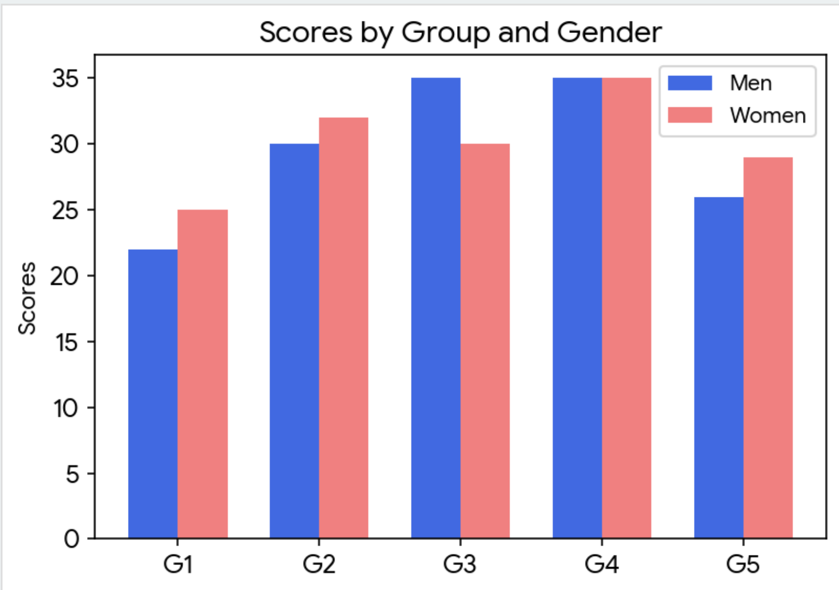
AIM

To write a Python program to create a bar plot of scores by group and gender, using multiple X values on the same chart.

CODE

```
import matplotlib.pyplot as plt
import numpy as np

groups = ['G1', 'G2', 'G3', 'G4', 'G5']
men_means = [22, 30, 35, 35, 26]
women_means = [25, 32, 30, 35, 29]

x = np.arange(len(groups))
width = 0.35

fig, ax = plt.subplots()
ax.bar(x - width/2, men_means, width, label='Men', color='royalblue')
ax.bar(x + width/2, women_means, width, label='Women', color='lightcoral')

ax.set_ylabel('Scores')
ax.set_title('Scores by Group and Gender')
ax.set_xticks(x)
ax.set_xticklabels(groups)
ax.legend()

plt.show()
```

OUTPUT



Result: The complex grouped bar chart comparing variables across multiple groups was successfully constructed.